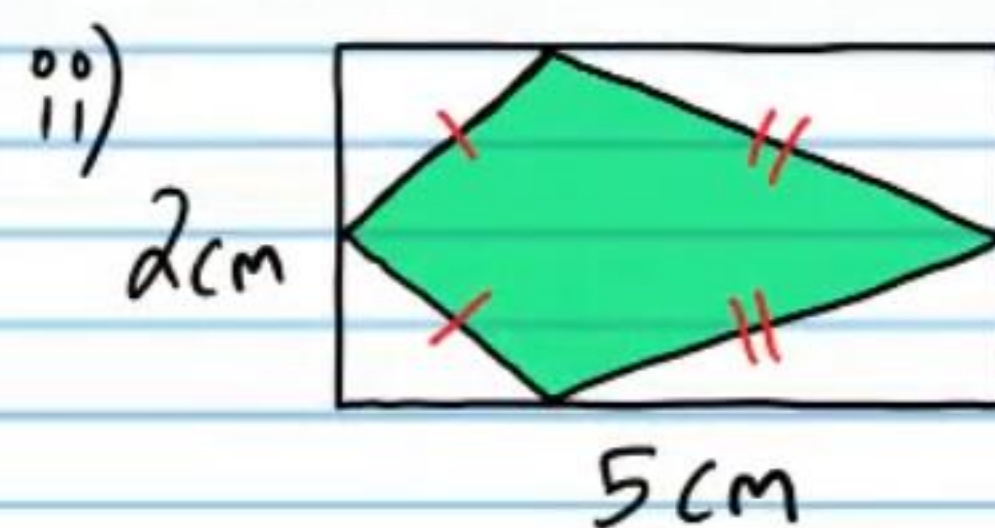
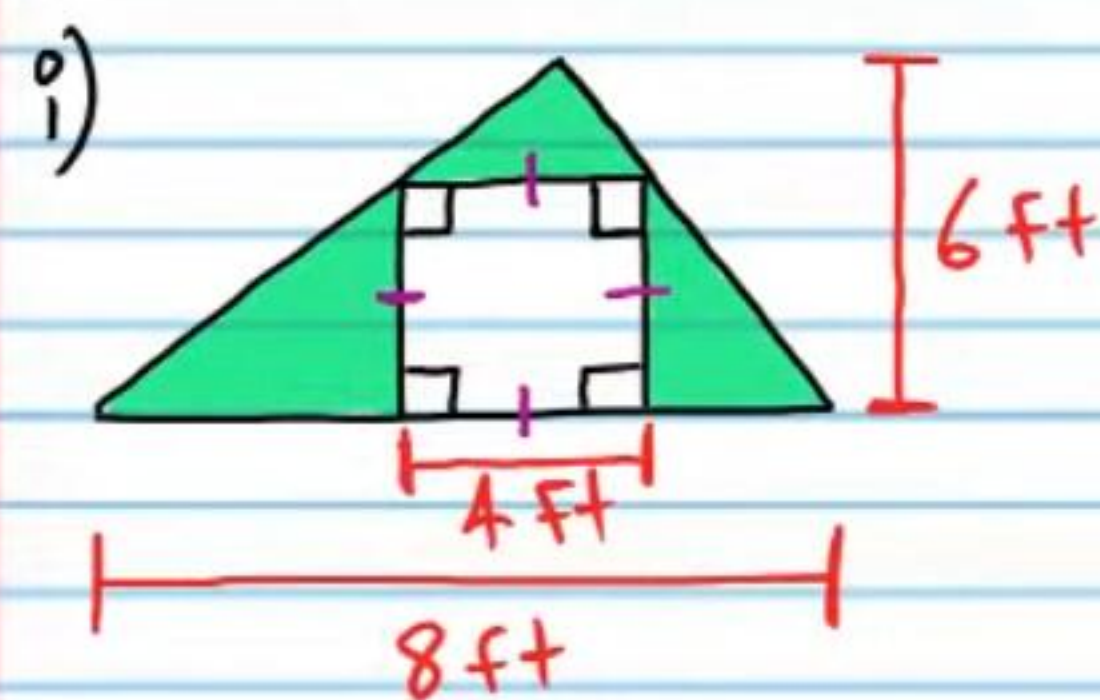


Practice Geometry Final Test-Part II (Version 2)

Name
Date
Course

- Write your name, the date, and the course title (i.e. Geometry).
- Show all work.
- You will need a calculator for some problems on this test. When pi is required, use the π button on your calculator.
- If a problem requires algebraic steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
- Graphing paper may be useful for this test.
- Each problem is worth four points for a total of 100 points.

① Find the area of each shaded region below.



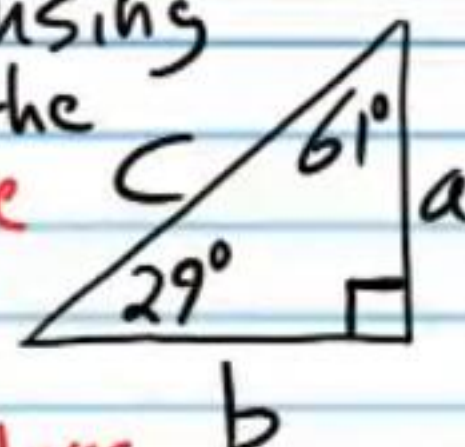
② What is the sum of the interior angles of a convex 20-gon?

③ Write the equation of a circle that has a diameter with endpoints $(-3, -5)$ and $(-1, -1)$.

④ i) Find the surface area of a rectangular prism with length 3 inches, height 2 inches, and width 6 inches.

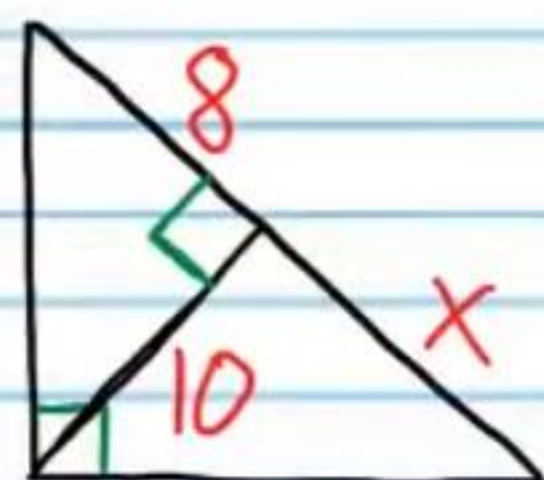
ii) Find the exact value of $\tan 30^\circ$ without a calculator. Hint: You must use a special triangle.

iii) Find $\cos 61^\circ$ using the triangle at the right. Do not use a calculator. Your answer should be a fraction with letters.

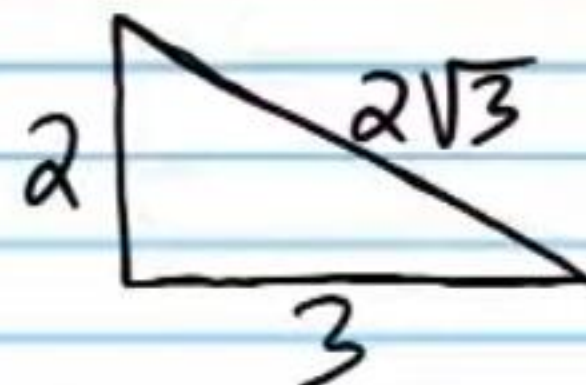


iv) Find $\cos 65^\circ$. Round to three decimal places.

5) i) Find X.



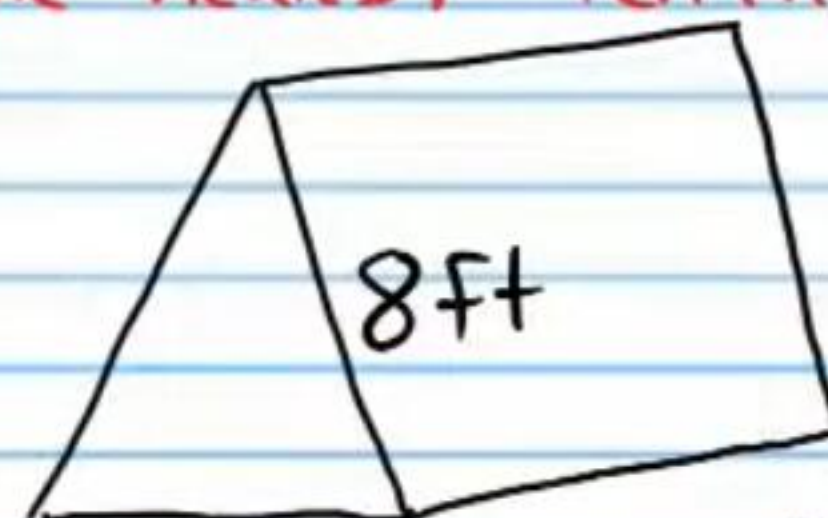
ii) Is the following polygon an acute, obtuse, or right triangle?



6) If the figures below are similar solids, find X. Round to the nearest tenth.

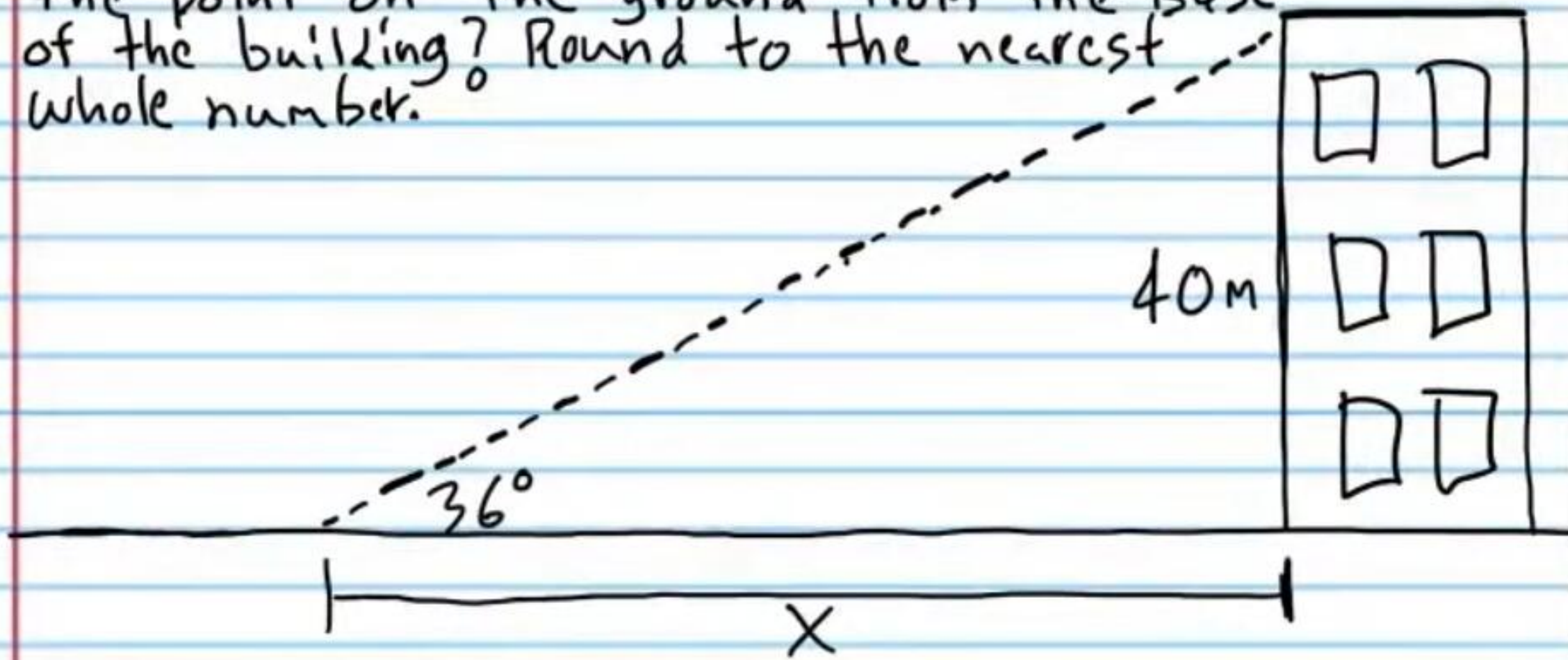


$$V_1 = 20 \text{ ft}^3$$



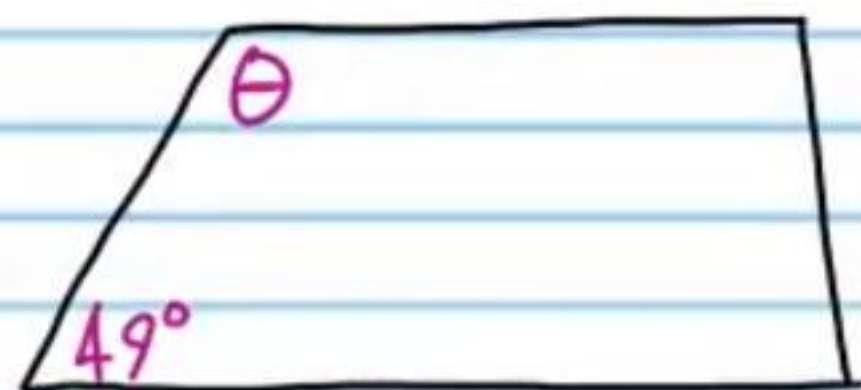
$$V_2 = 120 \text{ ft}^3$$

⑦ The angle of elevation from a point on the ground to the top of a building is 36° . If the building is 40 meters high, how far is the point on the ground from the base of the building? Round to the nearest whole number.



⑧ Find the unknown values in the trapezoids.

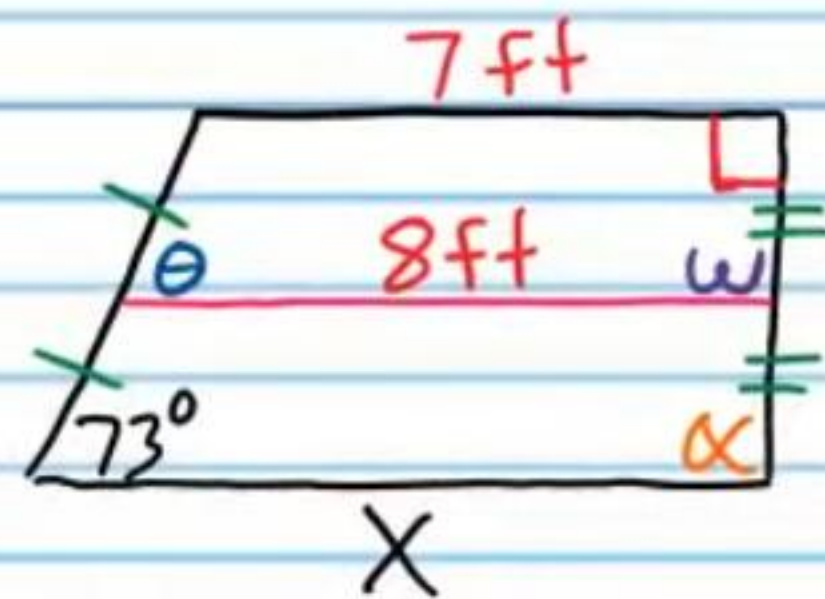
i) θ



ii) α

iii) w

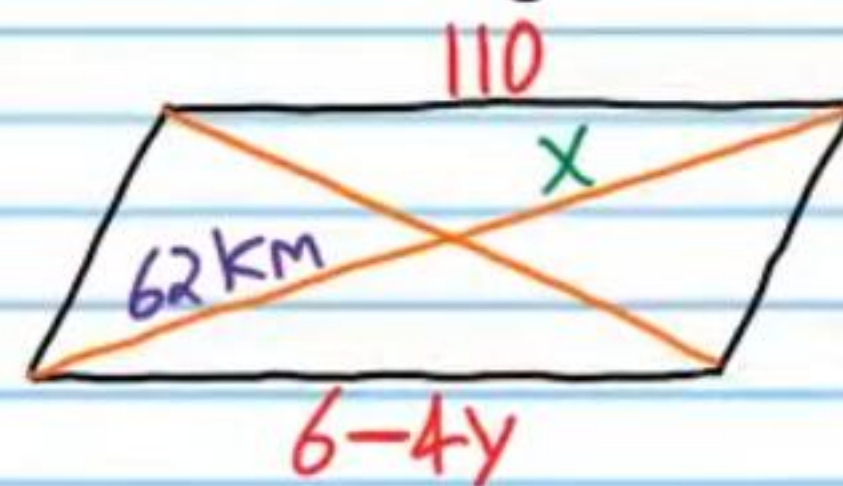
iv) x



⑨ Find the unknown values in the parallelogram.

i) y

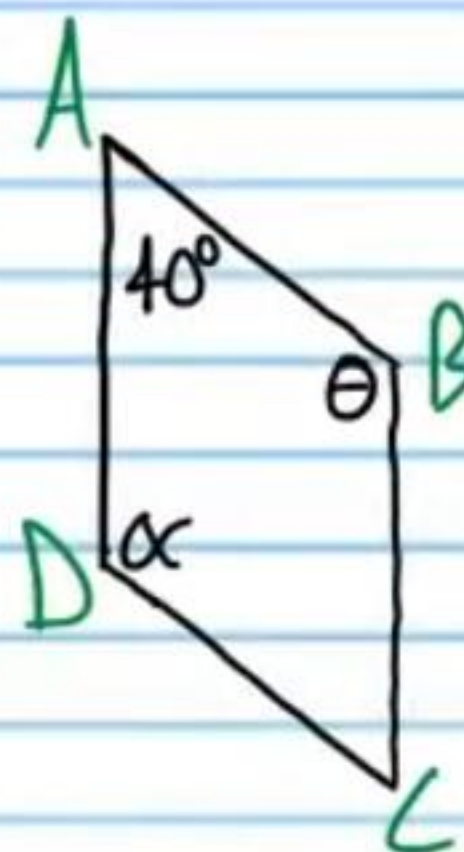
ii) x



Find the unknown values in the rhombus below.

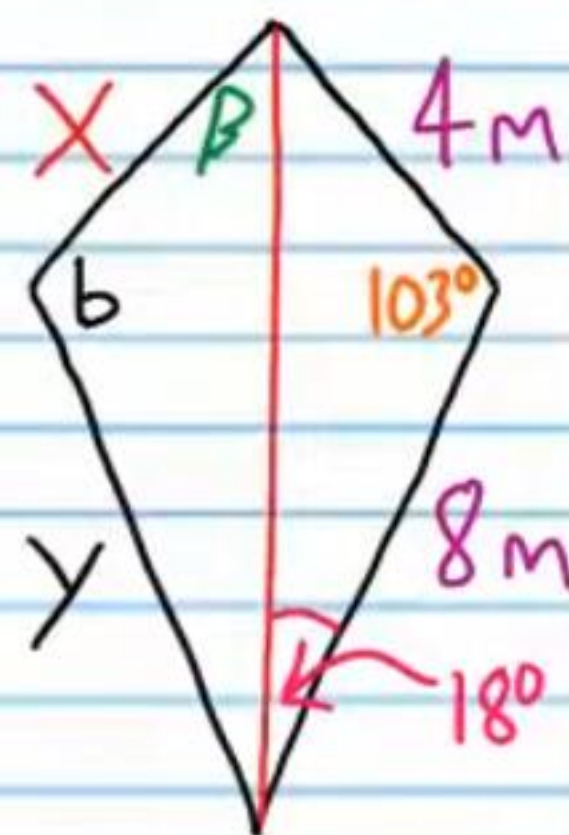
iii) α

iv) θ

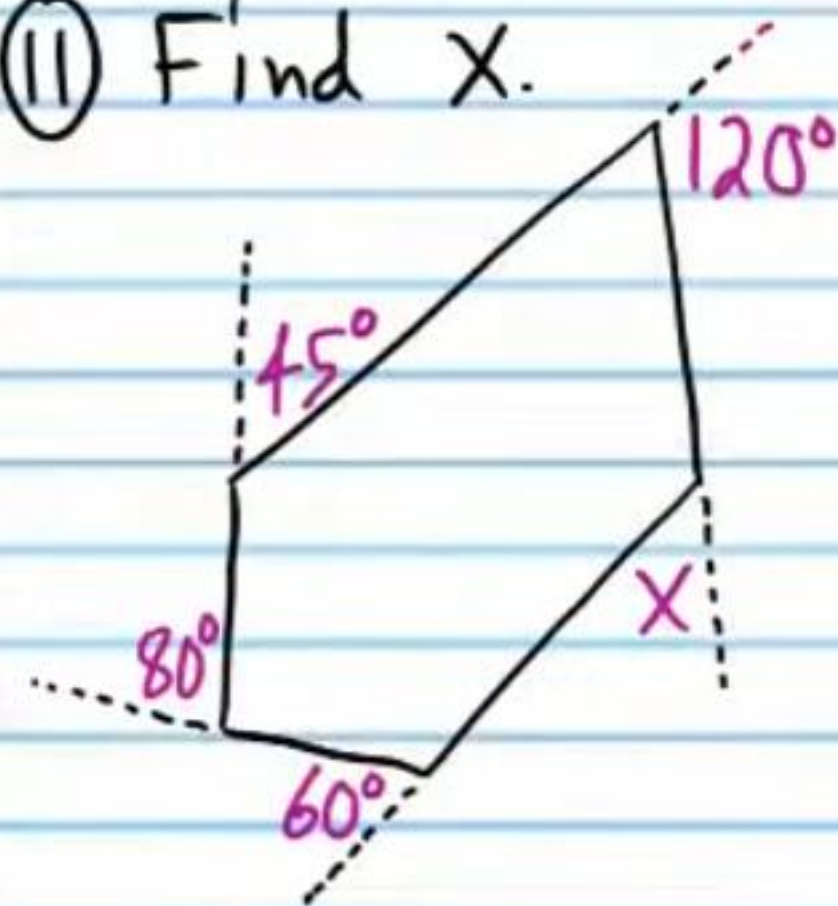


⑩ Find the unknown values in the kite below.

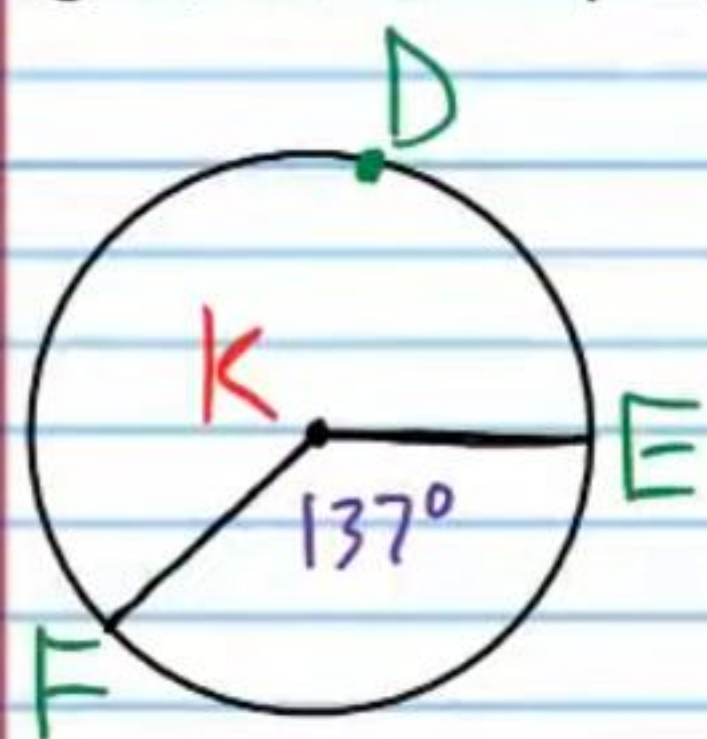
- i) y ii) x
 iii) β iv) b



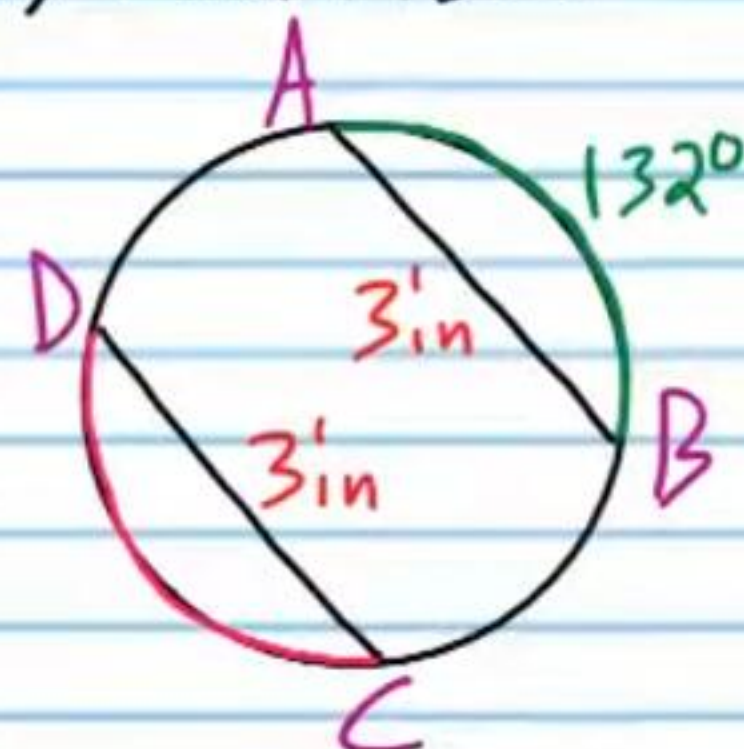
⑪ Find x .



⑫ i) Find $m\widehat{FDE}$ in $\odot K$.

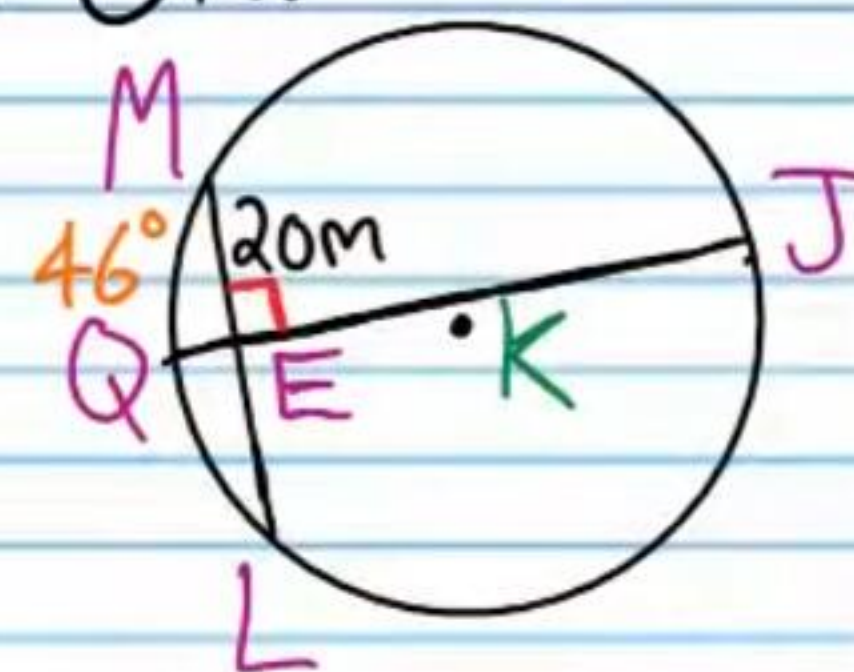


ii) Find $m\widehat{DC}$.



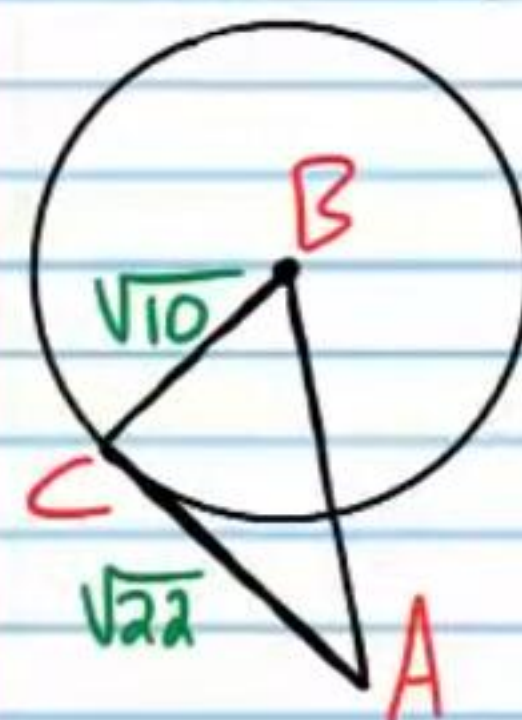
Find the following values in $\odot K$.

- iii) $m\widehat{ML}$
 iv) EL



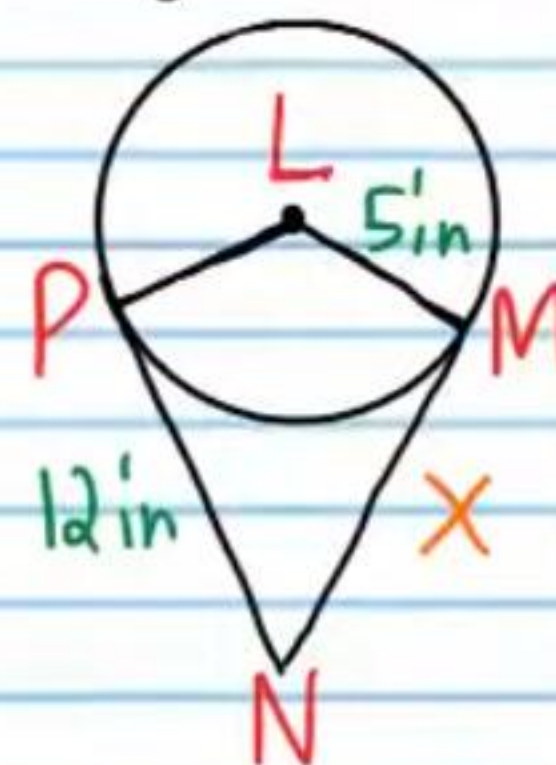
⑬ If $\overline{BC}$ is a radius, is the line containing $\overline{AC}$ tangent to $\odot B$? Justify your answer.

$$AB = 6$$



⑭ If the segments below are radii and tangents, find.

- i) LN ii) $m\angle P$

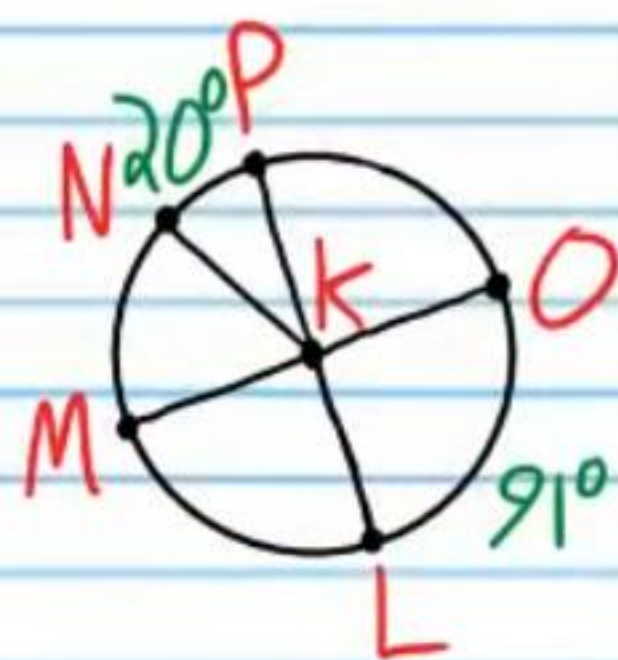


If $\overline{PL}$ and $\overline{MO}$ are diameters of $\odot K$ below, find

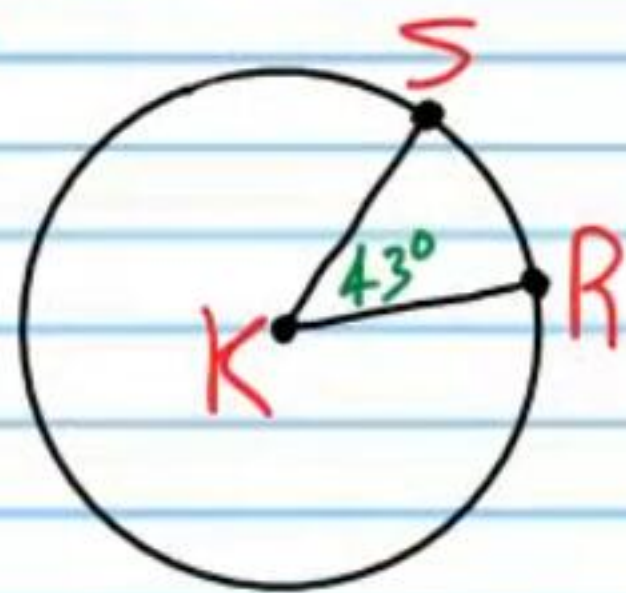
iii) $m\angle MKO$

iv) $m\widehat{NM}$

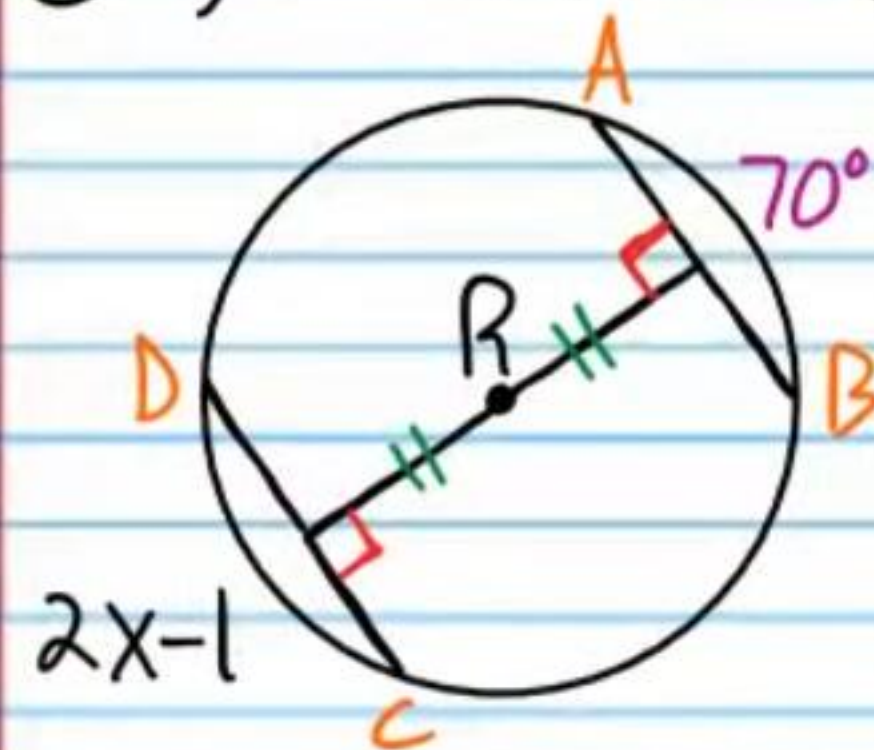
Hint: Use the vert. $\angle$'s thm.



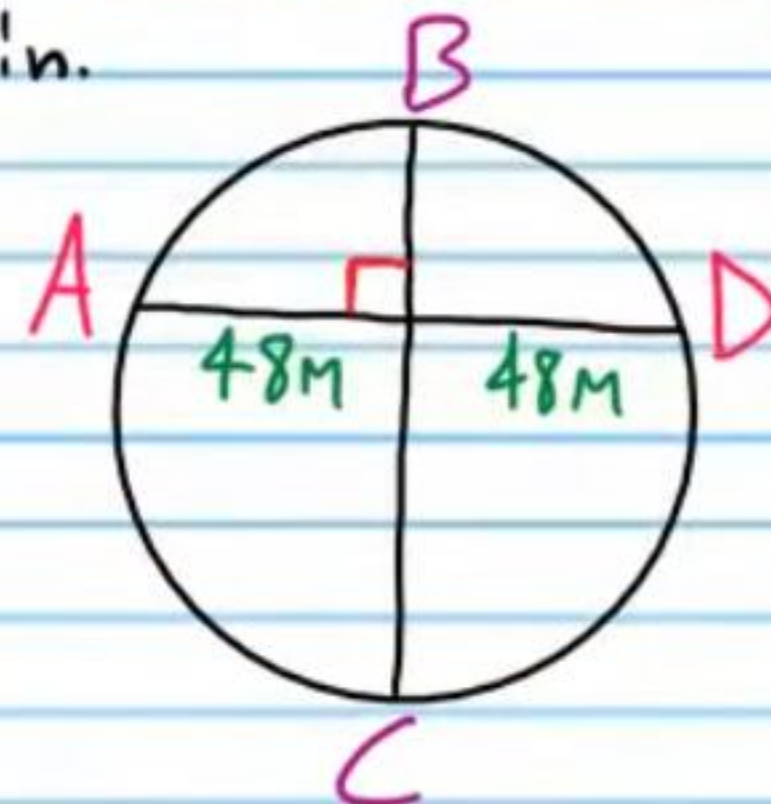
15) Find the diameter of $\odot K$ below if $\widehat{SR} = 5$ mm.
Round to the nearest hundredth.



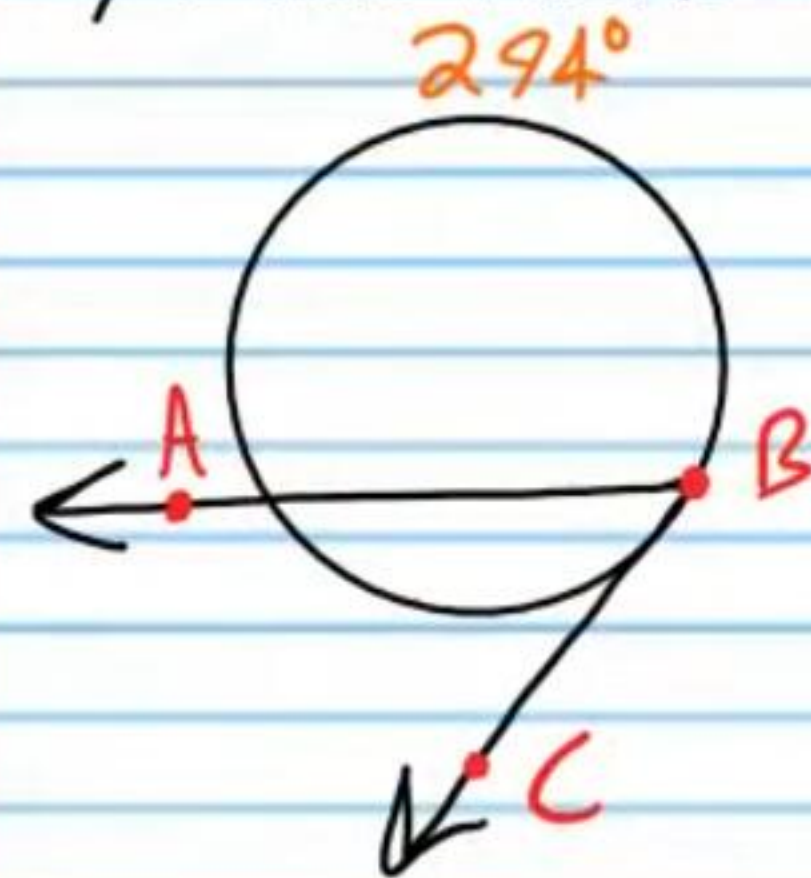
16) i) Find X in $\odot R$.



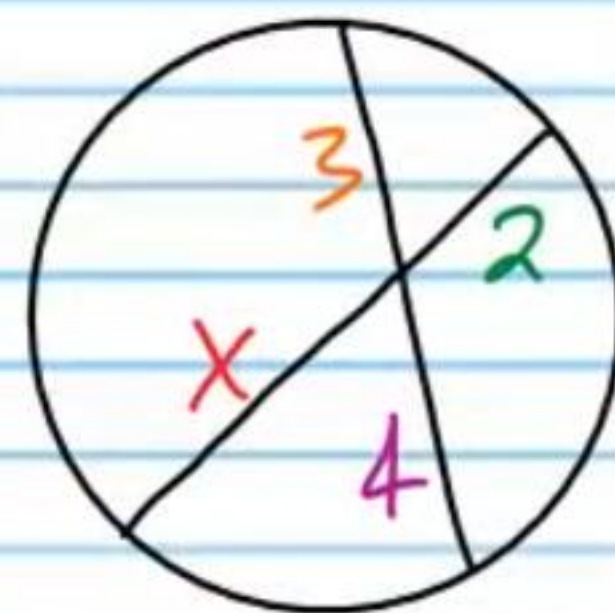
ii) Is $\overline{BC}$ a diameter? Explain.



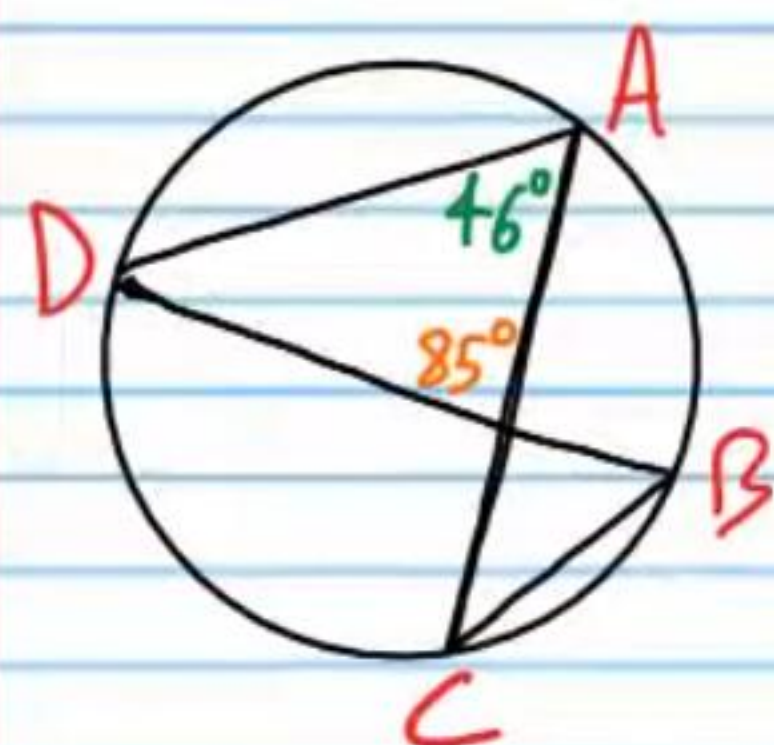
iii) Find $m\angle ABC$.



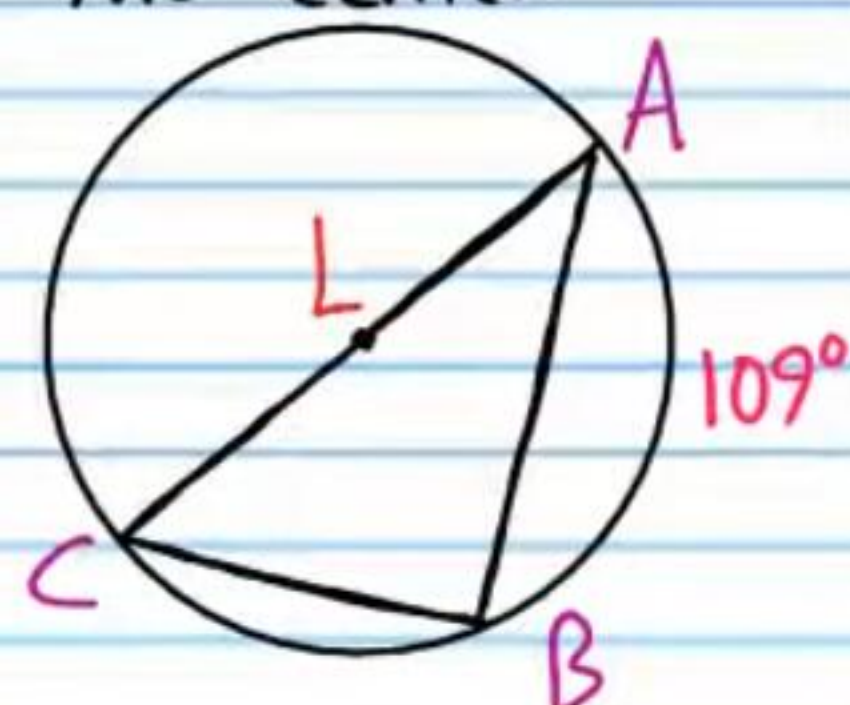
iv) Find X.



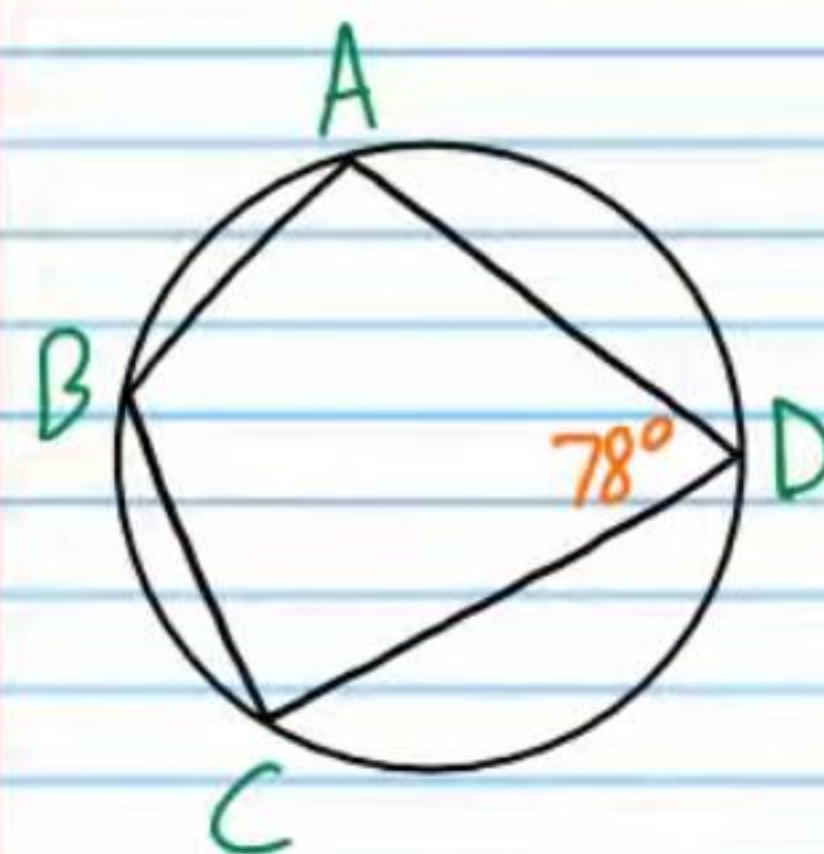
⑪ i) Find $m\angle ACB$.



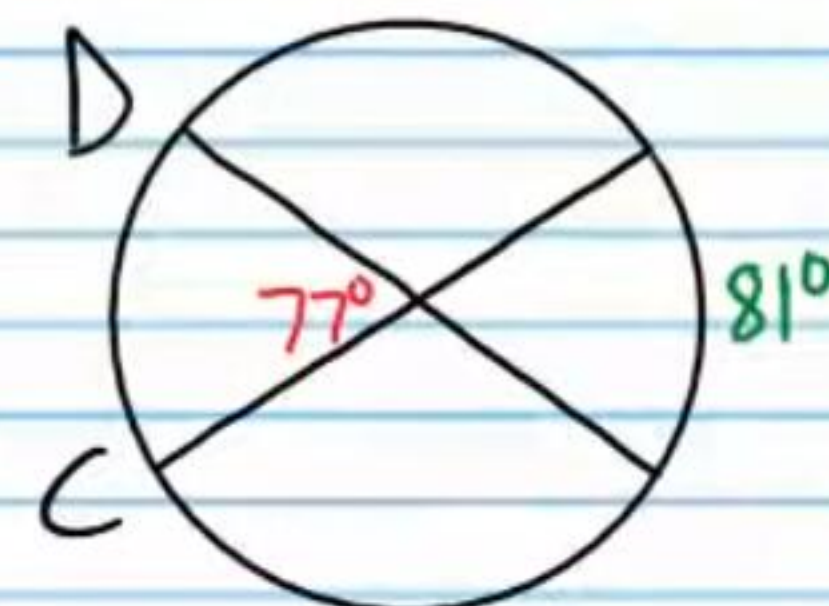
ii) Find $m\angle A$ if L is the center.



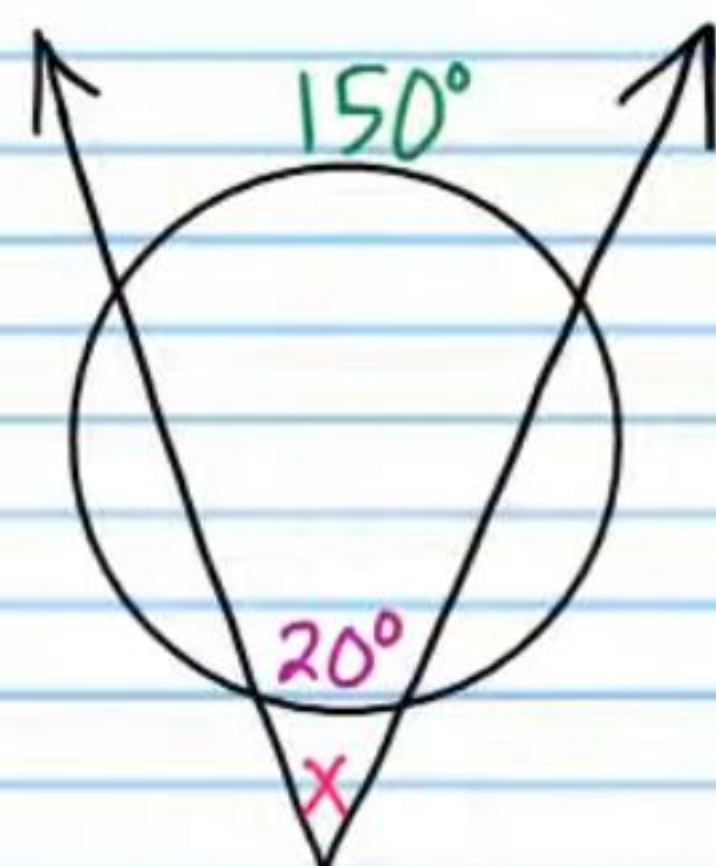
iii) Find $m\angle B$.



iv) Find $m\widehat{DC}$.



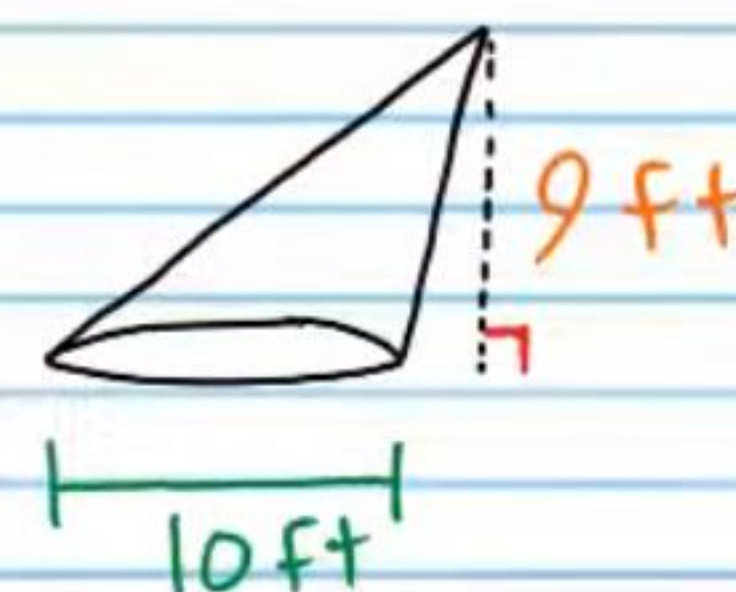
⑫ i) Find x .



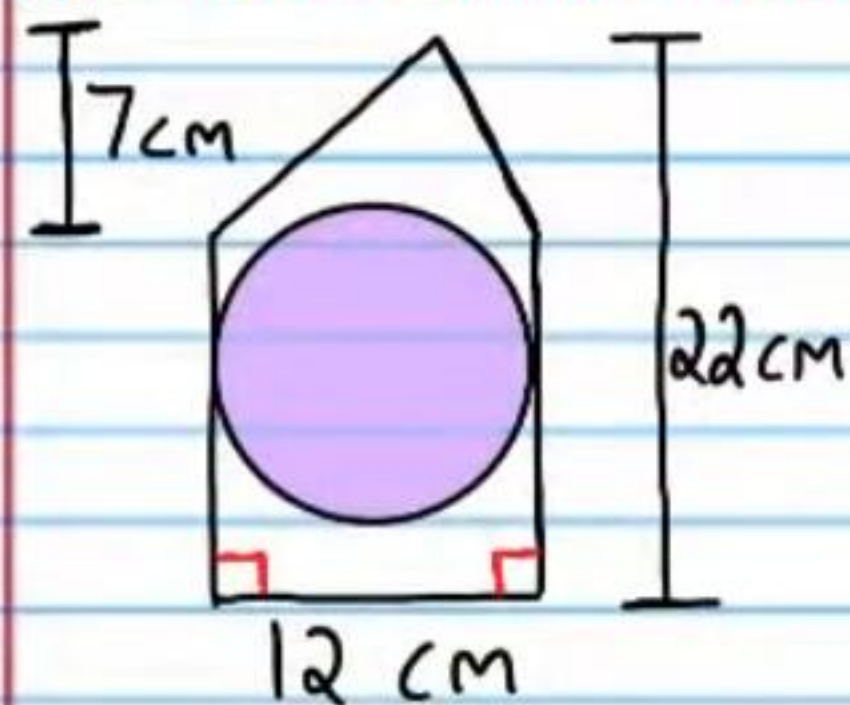
ii) Find y .



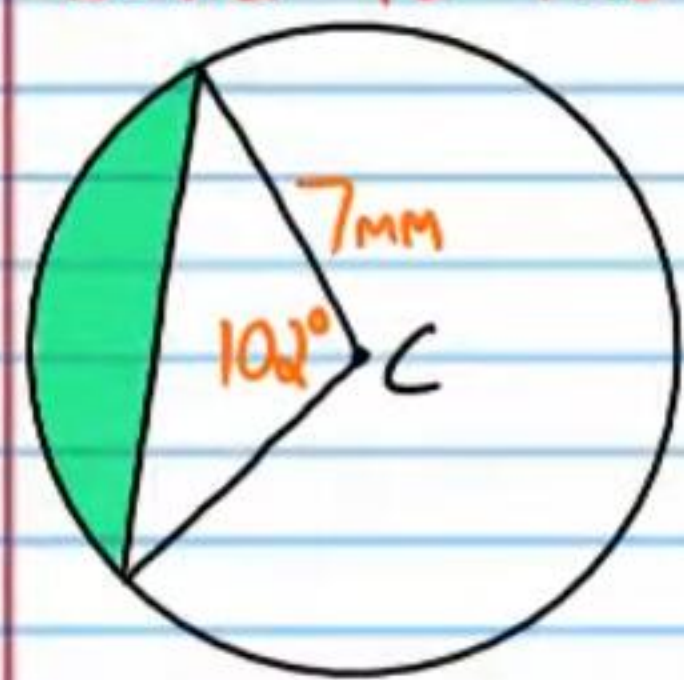
⑬ Find the Volume of the following solid S. Round to the nearest tenth.



- ②0 Find the probability that a randomly chosen point in the figure will lie in the shaded region. Write your answer as a fraction with a π symbol or as a decimal rounded to the nearest thousandth.

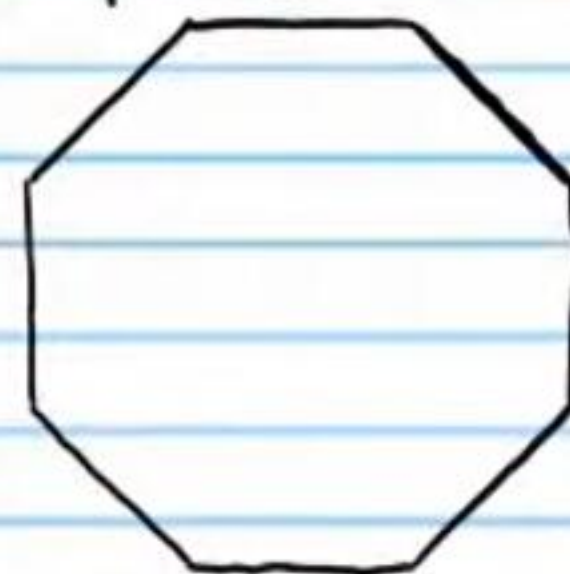


- ②1 Find the area of the shaded region in $\odot C$. Round to the nearest hundredth.



- ②2 Find the area of the regular polygon below. Round to the nearest tenth.

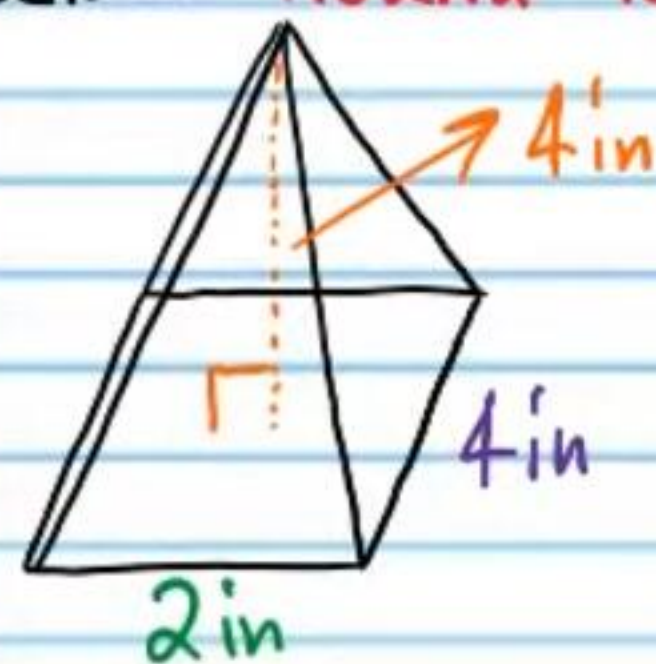
Apothem: 5 cm



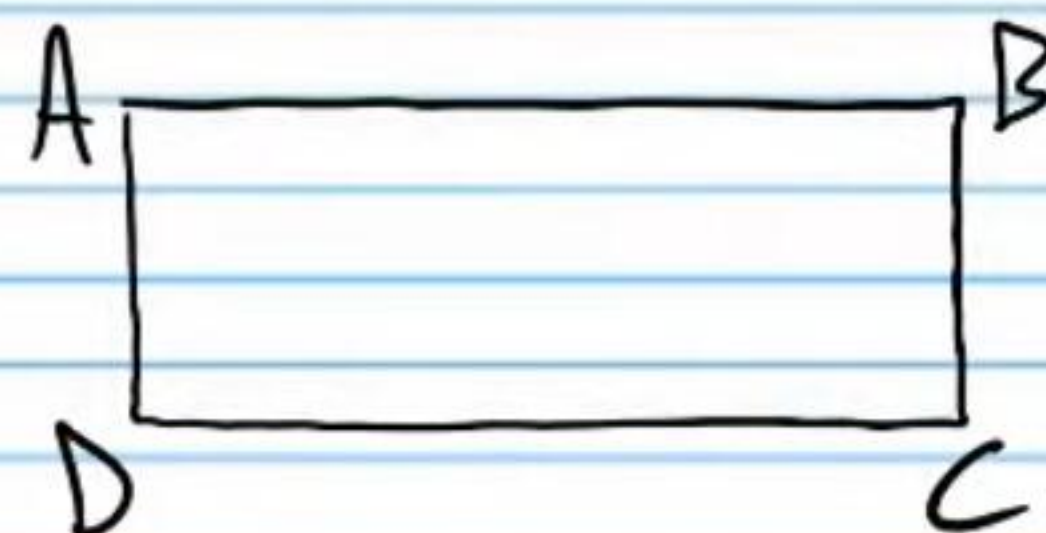
②③ Write a paragraph proof showing that if $H I J K$ below is a kite, then $\angle K H J \cong \angle I H J$ and $\angle K J H \cong \angle I J H$. *Hint: You must draw an auxiliary line.*



②④ Find the volume of the rectangular base pyramid below. Round to the nearest hundredth.



②⑤ Write a flowchart proof showing that if the quadrilateral below is a rectangle, then it is a parallelogram.



Practice Geometry Final Test-Part II (Version 2) Answers

① i) 8 ft^2

ii) 5 cm^2

② $3,240^\circ$

③ $(x+2)^2 + (x+3)^2 = 5$

④ i) 72 in^2

ii) $\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$

iii) $\frac{a}{c}$

iv) 0.423

⑤ i) $x = 12.5$

ii) Acute

⑥ $x = 4.4 \text{ ft}$

⑦ 55 m

⑧ i) $\theta = 131^\circ$

ii) $\alpha = 90^\circ$

iii) $w = 90^\circ$

iv) $x = 9 \text{ ft}$

⑨ i) $x = 62 \text{ km}$

ii) $y = -26$

iii) $\alpha = 140^\circ$

iv) $\theta = 140^\circ$

⑩ i) $y = 8 \text{ m}$

ii) $x = 4 \text{ m}$

iii) $\beta = 59^\circ$

iv) $b = 103^\circ$

⑪ $x = 55^\circ$

⑫ i) $m\widehat{FDE} = 223^\circ$

ii) $m\widehat{DC} = 132^\circ$

iii) Not enough information. $\overline{QJ}$ is not a diameter of $\odot K$ because it does not pass through K .

iv) Not enough information. $\overline{QJ}$ is not a diameter of $\odot K$ because it does not pass through K .

⑬ $(\sqrt{10})^2 + (\sqrt{22})^2 \neq 6^2$
 $10 + 22 \neq 36$
 $32 \neq 36$

No

⑭ i) $LN = 13 \text{ in}$

ii) $m\angle P = 90^\circ$

iii) $m\angle MKO = 180^\circ$

iv) $m\widehat{NM} = 71^\circ$

⑮ 13.32 mm

⑯ i) $x = 35.5$

ii) Yes. $\overline{BC}$ is a $\perp$ bisector of $\overline{AD}$.

iii) $m\angle ABC = 33^\circ$

iv) $x = 6$

⑰ i) $m\angle ACB = 49^\circ$

ii) $m\angle A = 35.5^\circ$

(18) i) $x = 65^\circ$

ii) $y = 2.5$

iii) $m\angle B = 102^\circ$

iv) $m\widehat{DC} = 73^\circ$

(19) i) $5,575.3 \text{ m}^3$

ii) 235.6 ft^3

(20) $\frac{18\pi}{111}$ or 0.509

(21) 19.65 mm^2

(22) 82.8 cm^2

(23) It is given that KHIJ is a kite. By the definition of a kite, $\overline{KH} \cong \overline{HI}$ and $\overline{KJ} \cong \overline{JI}$. Also, $\overline{HJ} \cong \overline{HJ}$ by the Reflexive Property of Congruence. Therefore, $\triangle KHJ \cong \triangle IJH$ by the SSS triangle congruence postulate. Finally, because corresponding parts of congruent triangles are congruent, $\angle KHJ \cong \angle IJH$ and $\angle KJH \cong \angle IJH$.

(24) 10.67 in^3

(25) ABCD is a $\square$.

Given

$\angle A, \angle D, \& \angle C$ are rt $\angle$'s.

★ You may use any group of three interior angles.

Def. of $\square$.

$m\angle A = 90^\circ, m\angle D = 90^\circ,$
and $m\angle C = 90^\circ$.

Def. of rt. $\angle$.

$m\angle A + m\angle D = 90 + 90$
& $m\angle D + m\angle C = 90 + 90$

Subs. Prop of $=$.

$m\angle A + m\angle D = 180^\circ$
 $m\angle D + m\angle C = 180^\circ$

Simplification.

$\angle A \& \angle D$ are supp. $\angle$'s
and $\angle D \& \angle C$ are supp. $\angle$'s.

Def. of Supp. $\angle$'s.



ABCD is a $\square$
Def. of a $\square$.

$\overline{AB} \parallel \overline{DC} \& \overline{AD} \parallel \overline{BC}$

CI ACT